

Experiment No 12

Self-induction and Mutual Induction

Objective

The objective of this experiment is to verify the phenomenon of self-induction and mutual induction.

Apparatus

- Digital Multi-Meter (DMM)
- Connecting Wires
- Self-Induction and Mutual Induction Trainer

Theory

Self-Induction

Any change in current causes an expansion or collapsing of the magnetic field around a conductor, which in turn induces an e.m.f. in the conductor. This phenomenon is self-induction induction. The magnitude of the self-induced e.m.f. is proportional to the amplitude and frequency of the current. By Lenz's Law, we can also find the relationship between the induced e.m.f. and the applied voltage that causes the current flow is such that the two voltages are always 180 degrees out of phase. Since the magnitude of the magnetic flux is proportional to the magnitude of the current in the coil. The self-induced e.m.f. of coil can be expressed by the Equation. Of course, the magnetic field of a coil can be made stronger, and more flux lines too, by keeping a soft iron core inside the coil.

$$e = L \frac{di}{dt}$$

Where L is the inductance of the coil due to which it opposes the change of current flowing through it. Inductance is attained by a coil due to the self-induced e.m.f produced in the coil itself by changing the current flowing through it. If the current in the coil is increasing, the self-induced e.m.f produced in the coil will oppose the rise of current, which means the direction of the induced e.m.f is opposite to the applied voltage. If the current in the coil is decreasing, the e.m.f induced in the coil is in such a direction as to oppose the fall of current; this means that the direction of the self-induced e.m.f is same as that of the applied voltage.

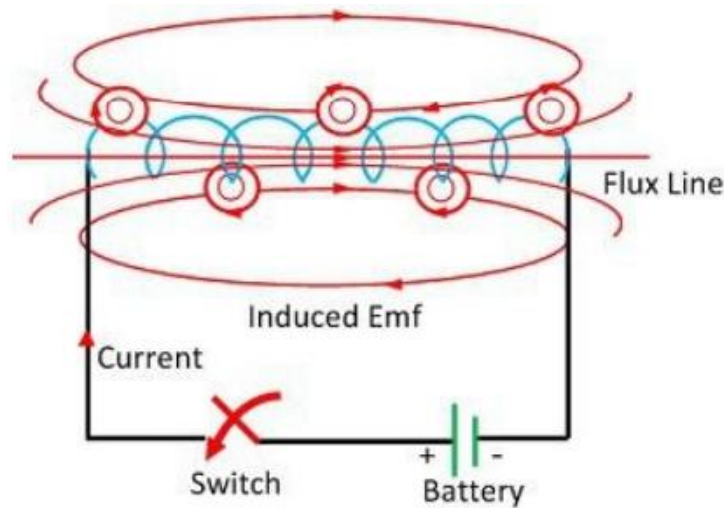


Figure 12.1: Self-Induction

This property of the coil only opposes the changing current (alternating current) and does not affect the steady current that is (direct current) when flows through it. The unit of inductance is Henry (H). The circuit of Figure 12.1 can be used to demonstrate the phenomenon of self-induction. When the switch is closed, the coil then current changes in the coil and builds a magnetic field (flux). If the switch is opened, the magnetic flux will collapse rapidly.

Mutual Induction

When the current in a coil is changing a voltage will be induced in another coil placed near to it. This effect is called mutual-induction. Consider Figure 12.2, current flowing through the coil of inductance L_1 builds up a magnetic field (flux) around itself. A part of this flux that links only coils 1. The other part links both the 1 and 2 coils and therefore is a component of mutual flux. This mutual flux sets up an induced e.m.f. in coil L_2 given by Equation.

$$e_m = M \left(\frac{dI_1}{dt} \right)$$

Or

$$M = e_m / \left(\frac{dI_1}{dt} \right)$$

Where, e_m is the voltage induced in the secondary coil and I_1 is the current flowing in the primary coil. M is the mutual inductance, defined as the property of the coils that enables it to oppose the changes in the current in another coil and measured in Henry (H). We can use this formula when we know the value of the mutually induced e.m.f as well as the change of current in coil two, or the neighboring coil. Now, if $e_m = 1$ and $dI_1 / dt = 1$, then on substituting the value in the given equations, we see that the value of M , that is mutual inductance is 1 Henry.

Thus two coils have a mutual inductance of 1 henry when e.m.f of 1 volt is induced in coil 1 and when the current flowing through coil 2 is changing at the rate of one ampere per second.

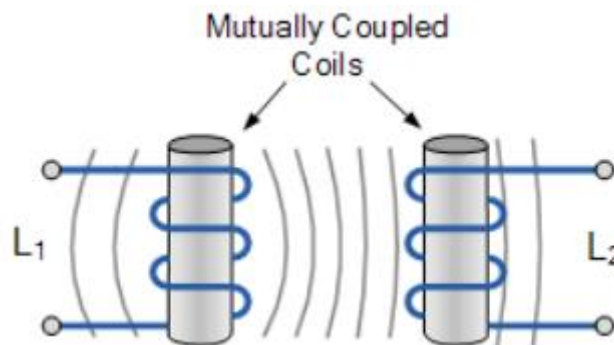


Figure 12.2: Mutual Induction

Procedure

1. Set the module for Self Induction.
2. According to Figure 6.3, complete the experiment circuit with short-circuit clips.
3. Observe the states of the lamp while pressing or releasing the switch and conclude your observation.
4. Set the module for the Mutual Induction.
5. Connecting the battery through a switch at one coil
6. Connect the other coil with the Galvanometer(G)
7. Observe the deflection on the galvanometer while pressing or releasing the switch and conclude your observation.

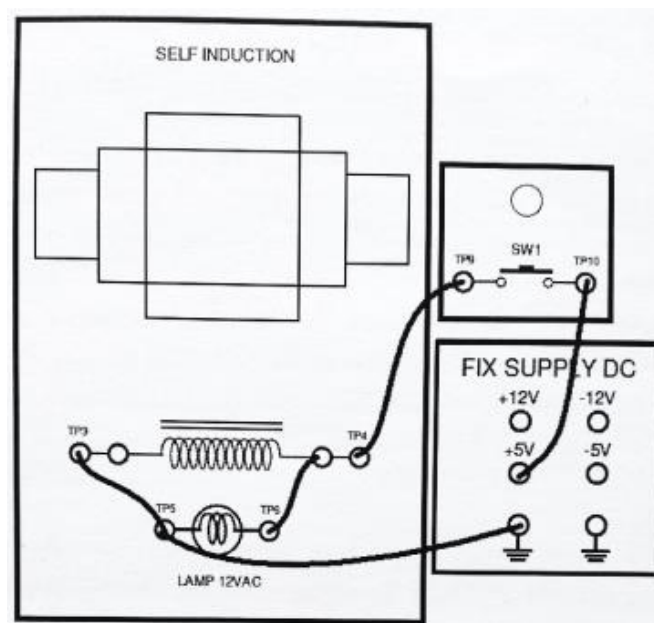


Figure 12.3: Connection Diagram of Self-Induction Experiment

[illegible]

LABORATORY SKILLS ASSESSMENT (Psychomotor)

Total Marks: 100

Criteria (Max Marks)	Level 1 $0\% \leq S < 50\%$	Level 2 $50\% \leq S < 70\%$	Level 3 $70\% \leq S < 90\%$	Level 4 $90\% \leq S \leq 100\%$	Score (S)
Procedural Awareness (20)	Selects inappropriate skills and/or strategies required by the task	Selects and applies appropriate skills and/or strategies required by the task with some errors	Selects and applies the appropriate strategies and/or skills specific to the task without significant errors	Selects and applies appropriate strategies and/or skills specific to the task without any error	
Practical Implementation (30)	Makes several critical errors in applying procedural knowledge of Self-induction and Mutual Induction	Makes few critical errors in applying procedural knowledge of Self-induction and Mutual Induction	Makes some non-critical errors in applying procedural knowledge Self-induction and Mutual Induction	Applies the procedural knowledge of Self-induction and Mutual Induction in perfect ways	
Safety (10)	Requires constant reminders to follow safety procedures	Requires some reminders to follow safety procedures	Follows safety procedures with only minimal reminders	Routinely follows safety procedures	
Use of Tool/Equipment (20)	Uses tools, equipment and materials with limited competence	Uses tools, equipment and materials with some competence	Uses tools, equipment and materials with considerable competence	Uses tools, equipment and materials with a high degree of competence	
Participation to Achieve Group Goals (10)	Shows little commitment to group goals and fails to perform assigned roles	Demonstrates commitment to group goals, but has difficulty performing assigned roles	Demonstrates commitment to group goals and carries out assigned roles effectively	Actively helps to identify group goals and works effectively to meet them in all roles assumed	
Interpersonal Skills in Group Work (10)	Rarely interacts positively within a group, even with prompting	Interacts with other group members if prompted	Interacts with all group members spontaneously	Interacts positively with all group members and encourages such interaction in others	
Marks Obtained					

Instructor's Signature: _____

Date: _____

LABORATORY SKILLS ASSESSMENT (Affective)

Total Marks: 40

Criteria (Max. Marks)	Level 1 $0\% \leq S < 50\%$	Level 2 $50\% \leq S < 70\%$	Level 3 $70\% \leq S < 90\%$	Level 4 $90\% \leq S \leq 100\%$	Score (S)
Introduction (5)	Very little background information provided or information is incorrect	Introduction is brief with some minor mistakes	Introduction is nearly complete, missing some minor points	Introduction complete and well-written; provides all necessary background principles for the experiment	
Procedure (5)	Many stages of the procedure are not entered on the lab report.	Many stages of the procedure are entered on the lab report.	The procedure could be more efficiently designed but most stages of the procedure are entered on the lab report.	The procedure is well designed and all stages of the procedure are entered on the lab report.	
Data Record (10)	Data is brief and missing significant pieces of information.	Data provides some significant information and has few critical mistakes.	Data is almost complete but has some minor mistakes.	Data is complete and relevant. Tables with units are provided. Graphs are labeled. All questions are answered correctly.	
Data Analysis (10)	Data is presented in very unclear manner. Error analysis is not included.	Data is presented in ways (charts, tables, graphs) that are not clear enough. Error analysis is included.	Data is presented in ways (charts, tables, graphs) that can be understood and interpreted. Error analysis is included.	Data are presented in ways (charts, tables, graphs) that best facilitate understanding and interpretation. Error analysis is included.	
Report Quality (10)	Report contains many errors.	Report is somewhat organized with some spelling or grammatical errors.	Report is well organized and cohesive but contains some grammatical errors.	Report is well organized and cohesive and contains no grammatical errors. Presentation seems polished.	
Marks Obtained					

LABORATORY SKILLS ASSESSMENT (Cognitive)

Total Marks: 10

(If any)	
Marks Obtained	

Instructor's Signature: _____

Date: _____